RESUME - Jamie Gabriel, PhD
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SUMMARY

- Passionate educator and learning designer with a strong background in classroom teaching, lesson planning, curriculum development, student mentoring and the design and delivery of educational workshops and programs. Experience working with primary, secondary and tertiary students across Music, History, Mathematics, Computer Science and STEM, including individual, small-group and classroom settings. Committed to engaging students of different ages, abilities and backgrounds through creative, practical and inclusive approaches to learning.
- Interdisciplinary academic, technology leader and researcher with expertise spanning computer science, data science, mathematics, music, educational technology and research. Doctoral-qualified in computational musicology, with a strong background in applying computational and analytical approaches to interdisciplinary research, teaching and practical problem-solving.
- Former Chief Data Officer at Cancer Institute NSW, with extensive experience in senior technology, data and research leadership across government, universities and public-sector organisations. Proven record of leading multidisciplinary teams, developing enterprise strategies, delivering complex technology initiatives and building collaborative relationships across research, education and professional environments.
- Co-founder of SAJA Music, an educational software company developing interactive software, digital learning environments and freely accessible resources for music students and teachers. Combines expertise in software engineering, artificial intelligence, music, mathematics and learning design to create engaging, accessible and practical approaches to music education.
- Currently undertaking a Master of Teaching at the University of Newcastle, specialising in Music and History, with NSW teacher accreditation. Building on existing teaching experience through formal studies in classroom practice, curriculum planning, assessment, differentiation, inclusive education and student engagement.
- Strong track record in curriculum development, educational research and innovation, particularly connecting emerging technologies and evidence-informed educational approaches with everyday classroom practice. Committed to developing inclusive, engaging learning experiences that accommodate diverse abilities and interests, and to contributing to curriculum consultation and the development of high-quality educational resources.

PROFESSIONAL EXPERIENCE

Co-founder – Educational Software & Learning Design
SAJA Music | January 2025 – Present

Co-founded SAJA Music, an educational software company developing interactive learning environments, digital applications and freely accessible educational resources for music students and teachers. Combines expertise in software engineering, artificial intelligence, music, mathematics and educational research to develop innovative approaches to teaching and learning.

- Lead the design and development of interactive educational software and web-based learning resources covering music theory, harmony, rhythm, aural skills, notation, composition and music technology.
- Develop educational applications incorporating real-time audio synthesis, interactive music notation, automated feedback and personalised learning activities.
- Apply contemporary software engineering, artificial intelligence and computational approaches to educational technology development, including the design, implementation, testing and deployment of web and mobile applications.
- Design and deliver individual and group music teaching, mentoring and practical learning activities, supporting students with diverse abilities, interests and learning needs.
- Develop accessible, technology-enhanced learning resources that encourage independent learning, creativity and engagement, with a commitment to making high-quality music education freely available.
- Undertake ongoing research and development exploring connections between music, mathematics, technology, pedagogy and curriculum.

Chief Data Officer
Cancer Institute NSW | December 2021 – August 2025

Senior executive responsible for enterprise data strategy, information governance, data architecture, analytics and digital transformation within a major public-sector health research organisation.

- Led the development and implementation of the Institute's enterprise Data Strategy, aligning information governance, technology architecture, analytics and research capabilities with organisational and NSW Health priorities.
- Directed multidisciplinary teams across data science, analytics, software engineering, information management and technology, supporting professional development, collaboration and organisational capability.
- Led major enterprise technology initiatives, including the development of cloud-based data infrastructure, research collaboration platforms and integrated information systems.
- Established enterprise data governance frameworks supporting responsible information management, privacy, security, research collaboration and appropriate use of sensitive health information.
- Developed collaborative partnerships across NSW Health, universities, research organisations and government agencies, supporting secure information sharing and research innovation.
- Provided strategic and technical advice to senior executives, translating complex research, technology and governance requirements into practical organisational solutions.

Manager, Data & Research Governance
Cancer Institute NSW | October 2017 – December 2021

Led the Institute's data and research governance functions, supporting the responsible management, analysis and application of health and research information.

- Developed and implemented enterprise frameworks for data governance, information management, privacy, security and responsible data use.
- Led the development of software systems and processes supporting data asset management, research governance, metadata, data lineage and information access.
- Established technical standards and collaborative development practices supporting data engineering, analytics, software quality, version control and research reproducibility.
- Developed systems for monitoring governance processes, information quality, risk and compliance, improving organisational accountability and operational efficiency.
- Worked closely with researchers, analysts, technology specialists and senior stakeholders to strengthen research and data capabilities across the organisation.

Lead, Data Projects
NSW Department of Education | 2015 – 2017

Led data, analytics and technology initiatives supporting education policy, planning, funding and organisational decision-making.

- Established data and analytics capabilities, governance processes and technical infrastructure supporting educational planning, reporting and policy development.
- Led multidisciplinary teams delivering enterprise software solutions, data integration projects, systems modernisation and automated reporting applications.
- Designed and implemented systems supporting the calculation and distribution of more than $200 million in NSW early childhood education funding, improving accuracy, accountability and operational efficiency.
- Developed collaborative data-sharing arrangements with NSW and Australian Government agencies, supporting secure information exchange and cross-agency analysis.
- Worked closely with education policy teams, executives and technical specialists to translate complex requirements into practical technology and information solutions.

SELECTED EARLIER ACADEMIC & PROFESSIONAL EXPERIENCE

Academic, Researcher & Educational Technology Developer
Macquarie University

Academic teaching, research and technology development across music, computing, learning analytics and interdisciplinary education.

- Delivered undergraduate and postgraduate teaching, lectures, tutorials and workshops across music, musicology, music history, harmony, improvisation, analysis and music technology.
- Designed and delivered creative arts and STEM education programs, including school outreach initiatives engaging primary and secondary students with diverse backgrounds and learning needs.
- Developed educational software, digital learning environments and research applications supporting teaching, student engagement and independent learning.
- Undertook research and development involving machine learning, predictive modelling, learning analytics and computational approaches to education and curriculum design.
- Led the development of university information systems and research applications, including technology supporting student administration, academic processes and learning analytics.
- Collaborated with academic staff, researchers, teachers and technical specialists on interdisciplinary research, teaching and educational technology initiatives.

General Manager, Faculty of Pharmacy
University of Sydney

Senior university leadership role encompassing organisational strategy, teaching and research infrastructure, technology development and academic operations.

- Led strategic initiatives supporting teaching, research, technology infrastructure and organisational development within a major university faculty.
- Managed the planning, development and implementation of teaching and research laboratories, coordinating academic, technical, professional and external stakeholders.
- Directed major technology and research computing initiatives, including the development of high-performance computing infrastructure supporting computational research.
- Designed and implemented enterprise information systems supporting scientific equipment management, research infrastructure and organisational operations.
- Worked collaboratively with academic and professional leadership to develop sustainable systems, infrastructure and operational processes supporting teaching and research.

EDUCATION & QUALIFICATIONS

Master of Teaching (Secondary)
University of Newcastle
Teaching Areas: Music and History
Currently completing

Doctor of Philosophy (PhD) – Computer Science / Computational Musicology
University of Technology Sydney
Research involving computational approaches to music analysis, machine learning and music information retrieval.

Bachelor of Arts (Honours Class I)
University of Sydney
History and Philosophy

Associate Diploma of Jazz Studies
Sydney Conservatorium of Music

PROFESSIONAL ACCREDITATION

- NSW Education Standards Authority (NESA): Conditional Accreditation to Teach
- NESA Accreditation Number: 347296
- NSW Working with Children Check: Current – Cleared for paid employment
- WWCC Number: WWC1119729E

TECHNICAL & EDUCATIONAL EXPERTISE

- Teaching & Learning: Classroom teaching, lesson planning, curriculum development, assessment, differentiation, inclusive education, student engagement, individual and group instruction, educational workshops and student mentoring.

- Educational Technology: Interactive learning environments, digital resource development, educational software, learning analytics, music notation, audio synthesis, automated feedback and AI-assisted learning tools.
- Software Engineering: Python, R, Haskell, C/C++, Objective-C, web and mobile application development, database design, systems architecture and software development lifecycle management.
- Data Science & Artificial Intelligence: Machine learning, statistical modelling, predictive analytics, data visualisation, computational research, information retrieval and applied AI.
- Research & Leadership: Interdisciplinary research, educational innovation, strategic planning, curriculum development, project leadership, stakeholder engagement, research governance and multidisciplinary team management.

SELECTED EDUCATIONAL TECHNOLOGY PROJECTS

SAJA Music – Interactive Learning Platform
Co-founder & Lead Developer | 2025–Present

Development of a freely accessible, web-based educational platform supporting music teaching, independent learning and the exploration of music through mathematics, computing and interactive technologies.

- Design and develop interactive learning resources covering music theory, aural skills, harmony, rhythm, notation, composition, piano technique and music technology.
- Develop browser-based educational applications incorporating real-time audio synthesis, interactive music notation, automated feedback and practical learning activities.
- Create structured learning pathways connecting educational resources and interactive exercises to support progressive skill development and independent learning.
- Apply software engineering, computational methods and learning design principles to develop accessible, engaging resources for students and teachers.

BeatMatrix – Rhythm Training Application (iOS & Web)
Developer | 2026

Designed and developed a standalone iOS application for rhythmic training, complex metres and tempo development.

- Developed an interactive rhythm practice environment supporting subdivisions, complex rhythmic structures and configurable tempo progression.
- Implemented flexible tempo transitions, rhythmic patterns and accurate audio scheduling to support technical practice and rhythmic understanding.
- Managed the complete software development lifecycle, including architecture, interface design, audio implementation, testing and deployment.
- Published the original iOS application through the Apple App Store and subsequently developed a browser-based version to support free and open access to educational technology.

REFEREES

Referees available on request.